

Business Analyst Working Test

King County Sheriff's Office

Time allotted: 1 hour

Background

You are supporting the King County Sheriff's Office Executive Leadership Team. The Sheriff has requested an initial review of Calls for Service to understand where workload is concentrated, how response performance varies, and whether service delivery differs across communities. The dataset you will use is anonymized and block-rounded; do not attempt to identify individuals.

This is a short exercise meant to show how you approach an analytical problem, make sense of messy operational data, and communicate early insights to leadership.

Materials Provided

You will receive access to a SharePoint file that includes:

- A three-year subset of Calls for Service data (in parquet format)
- Final Call Report (FCR) codes
- A Data Dictionary
- A Quarterly Business Review (QBR) Template

No additional geospatial sources are needed.

Your Assignment (60 minutes)

Step 1. Review the QBR Template

Begin by reading the template instructions. They explain both the content you need to produce and how your findings should be presented.

Step 2. Explore the Dataset

Using Python, R, or Excel (with visible formulas), examine the Calls for Service data to understand where call volume is concentrated by type, time, and geography; where response times are longest and how they differ across areas; and whether particular stages of the response (for example, travel time to scene) take longer than others. Summarize the most important patterns, what might be contributing to them, and what additional questions or follow-up analysis you would pursue.

Step 3. Produce One Metric and One Cleaning Process

In the same working file you used for your analysis, create one example KPI metric from the data and show one repeatable cleaning function or process. Your working file does not need to be polished, please leave your steps visible so your approach is clear.

Step 4. Build One Chart

Create one static chart or visual from your analysis and insert it into the QBR template. You may export or screen shot the chart from whatever tool you used.

Step 5. Complete the Notes Section of the QBR Template

Write your narrative using the standard Monthly Business Review structure:

- State plainly whether performance is on track relative to expectations.
- Identify the one or two factors most likely responsible for driving the trend you found.
- Describe root causes the department should explore.
- Propose specific, testable hypotheses for why those factors exist.
- Recommend one practical next step for leadership and note one limitation of this fast-turn analysis.

Step 6. Present your analysis at your interview

Evaluation Criteria

- **Technical execution:** correct use of the chosen tool, function/calculations, sound handling of data issues
- **Analytical thinking:** identifies meaningful signal quickly; recognizes limitations
- **Template narrative:** follows the on-track / driver / root cause / next-step structure
- **Business communication:** connects findings to operational decisions leadership can act on
- **Clarity:** clear, logical, concise executive writing; appropriate tone
- **Equity:** considers whether patterns differ across communities, without overreaching on causation